1.4 Matrix Group

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Matrix Group

Definition 0.1. In $\mathbb{C}^n$, $\overline{e}_i = (0, \dots, 1^{\text{th}}, 0, \dots)$ is a vector space basis element.

Definition 0.2. An *endomorphism* α of $\mathbb{C}^n$ is defined as

$$\alpha\overline{e}_i = \sum_{j=1}^{n} a_{ji}\overline{e}_j$$

where $\alpha = (a_{ij}) \in M_n(\mathbb{C})$. (We can think of $\alpha\overline{e}_i as the i^{\text{th}}$ column vector.)

We use α for a matrix as well as for the endomorphism.

We define multiplication of two endomorphisms α, β with corresponding matrices (a_{ij}), (b_{ij}) and construct (c_{ij}) where

$$c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

Lemma 0.1. $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$ as vector spaces.

Proof. Construct the function

$$f \colon M_n(\mathbb{C}) \to \mathbb{C}^{n^2},$$
$$(a_{ij}) \mapsto (b_{ij}),$$

where $b_{i+(j-1)n} = a_{ij}$.

Observe as j goes from 1 to n, $(j-1)n$ goes from 0 to $(n-1)n$ and as i goes from 1 to n as well, we get $i + (j-1)n$ go from 1 to n^2.

Observe f is surjective by construction and if $a_{ij} \neq a_{i'j'}$ then $b_{i+(j-1)n} \neq b_{i'+(j'-1)n}$ hence f is bijective.

Observe f is linear hence $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$. $\square$

Remark 0.1. Observe we can equip $M_n(\mathbb{C})$ with a topology by ensuring the given map above to be a homeomorphism. By constructing

$$\mathcal{T}_{M_n(\mathbb{C})} = \left\{ f^{-1}(U) \mid U \in \mathcal{T}_{\mathbb{C}^{n^2}} \right\}.$$

Lemma 0.2. Let T be any topological space,

$$\phi \colon T \to M_n(\mathbb{C}).$$

ϕ is continuous if and only if $\pi_{ij} \circ \phi$ is continuous.

Proof. Only if condition is trivial.

If condition:

$$\pi_{ij} \circ \phi \colon T \to \mathbb{C} \text{ is continuous.}$$

Now $\prod_{1 \le i,j \le n} \pi_{ij} \circ \phi \colon T \to \mathbb{C}^{n^2}$ is continuous by product topology (or box topology as $\mathbb{C}^{n^2}$ is finite Cartesian product). $\square$

Corollary 0.1. The product of two matrices is continuous.

Proof. Consider the map

$$q : M_n(\mathbb{C}) \times M_n(\mathbb{C}) \to M_n(\mathbb{C})$$

defined by matrix multiplication, i.e.,

$$q(\sigma, \tau) = \sigma\tau.$$

We can factor q as follows:

$$\begin{array}{ccc} M_n(\mathbb{C}) \times M_n(\mathbb{C}) & \xrightarrow{q} & M_n(\mathbb{C}) \\ \downarrow f \times f & & \downarrow f \\ \mathbb{C}^{n^2} \times \mathbb{C}^{n^2} & \xrightarrow{\rho} & \mathbb{C}^{n^2} \end{array}$$

where $f : M_n(\mathbb{C}) \to \mathbb{C}^{n^2}$ is the isomorphism identifying matrices with their entries, and ρ is a polynomial map defined by

$$\rho((b_{ij}), (b'_{ij})) = (c_{ij}),$$

with

$$c_{ij} = \sum_{k=1}^{n} b_{ik} b'_{kj}.$$

Since ρ is a polynomial map, it is continuous. The map f is a homeomorphism, and thus $f \times f$ is also continuous. Therefore, the composition $q = f \circ \rho \circ (f \times f)$ is continuous.

For any open set U in $M_n(\mathbb{C})$, we can express $U = f^{-1}(O)$ for some open set O in $\mathbb{C}^{n^2}$. Since q is continuous, the preimage $q^{-1}(U)$ is open in $M_n(\mathbb{C}) \times M_n(\mathbb{C})$. This proves the continuity of the matrix product. $\square$

$$q^{-1}(U) = q^{-1}(f^{-1}(O)) = (f \circ q)^{-1}(O)$$

which is open as $f \circ q$ is continuous.

$\square$

Notation: ${}^tA$ is the transpose of a matrix and $\overline{A}$ is the complex conjugate of a matrix.

Lemma 0.3. Transpose and complex conjugate are homeomorphisms.

Proof. If $h : M_n(\mathbb{C}) \to M_n(\mathbb{C})$ is defined by

$$A \mapsto {}^tA,$$

we can construct $r : \mathbb{C}^{n^2} \to \mathbb{C}^{n^2}$ which is a rearrangement of indices such that

$$b_{i+jn} = b_{j+in},$$

which is a homeomorphism. Hence, $h = f^{-1} \circ r \circ f$ is a homeomorphism.

If $k : M_n(\mathbb{C}) \to M_n(\mathbb{C})$ is homeomorphic by

$$A \mapsto \overline{A},$$

□

as k_{ij} is homeomorphic for $1 \leq i, j \leq n$.

Remark 0.2. Properties of matrices:

$$(\alpha\beta)^t = \beta^t\alpha^t$$
$$\frac{d(\alpha\beta)}{dt} = \dot{\alpha}\beta + \alpha\dot{\beta}$$

Definition 0.3. A regular (non-singular) $n \times n$ matrix, if it has an inverse, i.e., if there exists a matrix σ^{-1} such that

$$\sigma\sigma^{-1} = \sigma^{-1}\sigma = E,$$

where E is the unit matrix of degree n.

A necessary and sufficient condition for σ to be regular is $\det \sigma \neq 0$.

Remark 0.3. If σ is a regular matrix, then:

(i) iff σ^{-1} (the reciprocal endomorphism) exists;

(ii) then ${}^t(\sigma^{-1}) = ({}^t\sigma)^{-1}$, $(\overline{\sigma})^{-1} = (\overline{\sigma^{-1}})$.

Remark 0.4. If σ, τ are regular matrices, then $\sigma\tau$ are regular matrices.

Definition 0.4. The determinant map

$$\det : M_n(\mathbb{C}) \to \mathbb{C}$$

can be expressed as

$$\begin{array}{ccc} M_n(\mathbb{C}) & \xrightarrow{\det} & \mathbb{C} \\ \downarrow f & & \\ \mathbb{C}^{n^2} & \xrightarrow{p} & \mathbb{C} \end{array}$$

where p is some polynomial function of n^2 variables, which is continuous. We know f is continuous. Hence, the determinant map is continuous.

Remark 0.5.

$$GL_n(\mathbb{C}) = \{\sigma \in M_n(\mathbb{C}) \mid \det \sigma \neq 0\}$$
$$= M_n(\mathbb{C}) \setminus \overset{-1}{\det}\{0\}$$

is a group and an open subset as det is a continuous map.

(ii) For $\sigma = (a_{ij}) \in GL_n(\mathbb{C})$ the coefficient b_{ij} of σ^{-1} are given by

$$b_{ij} = \frac{1}{\det \sigma} A_{ji},$$

where A_{ji} is a polynomial of σ. Hence mapping $\sigma \mapsto \sigma^{-1}$ is continuous on $GL(n, \mathbb{C})$. Moreover, it's a homeomorphism.

Remark 0.6. If $\sigma \in GL_n(\mathbb{C})$, we define $\sigma^* = ({}^t\sigma)^{-1}$. Then we have $(\sigma\tau)^* = \sigma^*\tau^*$. Hence $\sigma \mapsto \sigma^*$ is a homeomorphism, and an automorphism of order 2 of $GL_n(\mathbb{C})$.

Definition 0.5. Let $\mathrm{O}_n(\mathbb{C}) = \{\sigma \in \mathrm{GL}_n(\mathbb{C}) \mid \sigma^* = \sigma^{-1}\}$ be the set of orthogonal matrices. If $\sigma = \sigma^*$, then σ is called *complex orthogonal.* If $\sigma = \sigma^* = \overline{\sigma}$, then σ is called *real orthogonal.*

Definition 0.6. Let $\mathrm{U}_n(\mathbb{C}) = \{\sigma \in \mathrm{GL}_n(\mathbb{C}) \mid \sigma^* = \overline{\sigma}^{-1}\}$ be the set of all unitary matrices.

Remark 0.7. Note that $\mathrm{U}_n(\mathbb{C})$ or $\mathrm{O}_n(\mathbb{C})$ are non-empty as $e \in \mathrm{U}_n(\mathbb{C})$ and $e \in \mathrm{O}_n(\mathbb{C})$, where e denotes the identity matrix.

Remark 0.8. Notation: $\mathrm{U}_n(\mathbb{C}) = \mathrm{U}(n)$ and $\mathrm{O}_n(\mathbb{R}) = \mathrm{O}(n)$. Do not be confused!

Lemma 0.4. If f, g are continuous mappings from topological space X to Hausdorff space Y, then $\{x \in X \mid f(x) = g(x)\}$ is closed.

Proof. Consider $\{x \mid f(x) \neq g(x)\}$. Now $x \in \{x \in X \mid f(x) \neq g(x)\}$ implies $f(x) \neq g(x)$ in Y. By Hausdorff property there exists open set U_1 and U_2 such that $f(x) \in U_1$ and $g(x) \in U_2$ and $U_1 \cap U_2 = \emptyset$. Now $f^{-1}(U_1)$ and $g^{-1}(U_2)$ are open. Let $V = f^{-1}(U_1) \cap g^{-1}(U_2)$ be a nbd of x. For any $y \in V$, $f(y) \neq g(y)$ otherwise $U_1 \cap U_2 \neq \emptyset$. Hence $x \in V \subseteq \{x \mid f(x) \neq g(x)\}$ is open. Hence $\{x \mid f(x) = g(x)\}$ is closed. □

Proposition 0.1. $O_n(\mathbb{C})$, $U(n)$, $O(n)$ are closed subgroups of $GL_n(\mathbb{C})$.

Proof. Observe that the mappings $\sigma \mapsto \sigma^*$, $\sigma \mapsto \overline{\sigma}$, and the identity mapping are continuous.

By the above lemma, as $\mathbb{R}$ is a T_2 space, the sets

$$U(n) = \{\sigma \in GL_n(\mathbb{C}) \mid \sigma = \sigma^*\overline{\sigma}\}$$

and

$$O_n(\mathbb{C}) = \{\sigma \in GL_n(\mathbb{C}) \mid \sigma = \sigma^*\overline{\sigma}^T\}$$

are closed sets. Observe that

$$GL_n(\mathbb{R}) = \{\sigma \in GL_n(\mathbb{C}) \mid \sigma = \overline{\sigma}\}$$

is also closed. Hence $U(n)$ and

$$O(n) = O_n(\mathbb{C}) \cap GL_n(\mathbb{R})$$

is closed.

As all the given sets are non-empty by subgroup test, we can show these are subgroups. □

Definition 0.7. The special linear group is defined as

$$SL_n(\mathbb{C}) = \{\sigma \in GL_n(\mathbb{C}) \mid \det \sigma = 1\}.$$

Proof. As $\sigma \in SL_n(\mathbb{C})$, and by subgroup tests, we can show $SL_n(\mathbb{C})$ is a subgroup of $GL_n(\mathbb{C})$. Moreover, $SL_n(\mathbb{C}) = \det^{-1}\{1\}$ implies $SL_n(\mathbb{C})$ is a closed subgroup. $\square$

Remark 0.9.

$$SL(n) = GL_n(\mathbb{R}) \cap SL_n(\mathbb{C}),$$
$$SU(n) = SL_n(\mathbb{C}) \cap U(n),$$
$$SO(n) = SL_n(\mathbb{C}) \cap O(n)$$

are closed subgroups of $GL_n(\mathbb{C})$.

Theorem 0.1. *$U(n)$, $O(n)$, $SU(n)$ and $SO(n)$ are compact.*

Proof. Observe $O(n)$, $SU(n)$ and $SO(n)$ are closed subgroups of $U(n)$. It is sufficient to show $U(n)$ is compact.

Goal: $U(n)$ is homeomorphic to a closed and bounded set of $\mathbb{C}^{n^2}$.

Observe $GL_n(\mathbb{C})$ is an open subgroup of $M_n(\mathbb{C})$, hence it is also a closed subgroup according to a previous lemma. Observe $U(n)$ is closed in $GL_n(\mathbb{C})$ and $GL_n(\mathbb{C})$ is closed in $M_n(\mathbb{C})$ implies $U(n)$ is closed in $GL_n(\mathbb{C})$.

Now for $\sigma = (a_{ij}) \in U(n)$ if and only if

$${}^t\overline{\sigma}\sigma = \varepsilon \iff \sum_k a_{ki}\overline{a_{kj}} = \delta_{ij}.$$

$\square$

$$\sum_{k=1}^{n} |a_{ki}|^2 = 1$$

implies that for the i^{th} column vector, the absolute sum is 1. This implies for any $1 \leq i, j \leq n$, $|a_{ji}| \leq 1$.

Now in $(\mathbb{C}^{n^2}, \|\cdot\|_1)$, let $v = (v_k)_{k=1}^{n^2}$ where $|v_k| \leq 1$ for $1 \leq k \leq n^2$. We define

$$\|v\|_1 = \max_{1 \leq k \leq n^2} |v_k|.$$

Observe $f : M_n(\mathbb{C}) \to \mathbb{C}^{n^2}$ as a homeomorphism. $f(U(n))$ is closed and bounded, and compact. This implies $U(n)$ is compact.

$\square$

Remark 0.10. The choice of the norm does not matter as all norms in finite dimension vector space are equivalent (add the proof to that in the appendix).